5634080

B.Tech. DEGREE EXAMINATION, JULY 2023.

Fourth Semester

Computer Science Engineering

MICROPROCESSORS AND MICROCONTROLLERS

(2013-14 Regulations)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. List the registers of 8085 Microprocessor.

2. State any four pins of 8086 processor which are used to generate control and status signals.

3. Differentiate between Hardware and Software Interrupts.

4. How the DMA operations perform in Microprocessor?

5. What is Memory Mapped I/O?

6. Write down the signals available for Serial communication.

If the execution unit generates effective address of 43A2 H and the DS register contains 4000 H. What will be the physical address generated by the BIU? What is the Maximum Size of the data segment?

8. What is the addressing mode of MOV AX, 55H (BX) (SI)?

9. List any four features of 8051 microcontroller.

10. How can 8051 be interrupted?

PART B — (5 × 11 = 55 marks)

Answer ALL the questions.

UNIT I

11. (a) Analyze the Compare Instructions of 8085 Micro Processor. (4)

(b) Draw the timing diagram of MOV A, M instruction and explain each machine cycle. (7)

Or

12. Write an assembly language program to sort numbers in ascending order.

UNIT II

13. Explicate the silent features of 8259 and explain the block diagram of 8259 Programmable Interrupt Controllers.

Or

14. Summarize about "Synchronous and Asynchronous Serial Transmission".

UNIT III

15. With a neat sketch, explicate the components of 8279 Programmable Keyboard/display Interface and their functions.

Or

16. Enumerate the various classification of Memory and describe its functioning in detail.

UNIT IV

17. (a) Illustrate the pin diagram of 8086 Microprocessor. (5)

(b) Explain the following Assembler directives in 8086. (6)

(i) ASSUME.

(ii) EQU.

(iii) DW.

(iv) DD

Or

18. Describe the instruction set of 8086 in detail.

UNIT V

19. Discuss about the organization of Internal RAM and Special Function Registers of 8051 Microcontroller in detail.

Or

20. (a) Examine the Data transfer instructions and Program control instructions of 8051 micro controller. (6)

(b) Write an assembly language program based on 8051 microcontroller instruction set to perform four arithmetic operations on 2, 8 bit data. (5)

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5634080

B.Tech. DEGREE EXAMINATION,
SEPTEMBER 2021.

Fourth Semester

Computer Science and Engineering

MICROPROCESSORS AND MICROCONTROLLERS

Time : Three hours | Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. What is the need for ALE signal in 8085 microprocessor?

2. Specify the contents of the registers and the flag status, when the following instructions are executed.

   (a) MVI A, 00H

   (b) MVI B, F8H

   (c) MOVC, A

   (d) MOVD, B

   (e) HLT.

3. Differentiate between burst mode data transfer and cycle stealing method of data transfer in DMA.

4. List the 8085 interrupts based on priority, maskable or non-maskable. vectored on non-vectored.

5. What are the advantages of using the USART chips in microprocessor based systems?

6. Compare memory mapped I/O and programmed I/O technique. Differentiate blocking and non-blocking assignments in Verilog.

7. What are the three classifications of 8086 interrupts?

8. State the significance of HOLD and LOCK signals in 8086. Mention the various system tasks available in Verilog with suitable examples.

9. Compare the features of microprocessor and microcontroller.

10. What is the function of GATE bit in the TMOD register of 8051?

UNIT V

19. (a) Describe how memory is organized in 8051 based system. (6)

(b) Add the unsigned number found in 8051 microcontroller internal RAM locations 25h, 26h and 27h together and put the result in RAM locations 30h (MSB) and 31h. (5)

Or

20. (a) Bring out the features of Special Function Registers of 8051 microcontroller. (6)

(b) Describe the 8051 I/O port structure. (5)

PART B — (5 × 11 = 55 marks)

Answer ALL questions, One from each Unit.

All questions carry equal marks.

UNIT I

11. (a) With the help of neat diagram explain the architecture of 8085 microprocessor in detail. (7)

(b) Draw and explain the timing diagram of opcode fetch cycle. (4)

Or

12. (a) With suitable examples explain 8085 addressing modes in detail. (7)

(b) An array of bytes is stored starting from memory location, C301H. Length of this array is stored in memory location C300H. Count how many bytes are greater than 70H. Store this count in memory location C400H. (4)

UNIT II

13. Explain different modes of 8253 programmable interval timer in detail.

Or

14. Explain with block diagram the operation of 8259A programmable interrupt controller in detail. Explain control word definition of the same.

UNIT III

15. (a) Design an interface circuit needed to connect DIP switch as an input device and display the value of the key pressed using a 7 segment LED display. Using 8085 system, write a program to implement the same. (7)

(b) Write short notes on RS232 interface. (4)

Or

16. (a) With schematic explain the role of 8255 in I/O interfacing. (6)

(b) Design a microprocessor system to interface an 8K × 8 EPROM and 8K × 8 RAM. Give the detailed memory map and address decoding logic. (5)

UNIT IV

17. (a) Explain the instruction set of 8086 microprocessor. (8)

(b) Give the functions of NMI, BHE and TEST pins of 8086. (3)

Or

18. (a) Explain the maximum mode configuration of 8086 with a neat diagram and explain the signals (6)

(b) Explain how the memory unit is addressed by 8086 with a neat diagram. (5)

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5634080

B.Tech. DEGREE EXAMINATION, SEPTEMBER 2020.

Fourth Semester

Computer Science and Engineering

MICROPROCESSORS AND MICROCONTROLLERS

(2013-14 Regulations)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Compare microprocessor and microcontroller.
2. Explain Flag Register and draw its format.
3. List the programmable registers in 8085 processor.
4. What is DMA?
5. What is USART?
6. Write a note on RS232C interface.
7. What is Interrupt Processing?

8. List the addressing modes of 8086.

9. Enlist the timer mode 8051 microcontroller.

10. Calculate baud rate if 8051 microcontroller as frequency = 12 MHz, SMOD = 1 and reload value = FFH.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL the questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the functional blocks of 8085 microprocessor with a neat block diagram.

Or

12. Write an assembly language program to convert two digit BCD number stored in memory into a Hexa Decimal Number. Use the NEAR procedure call.

UNIT II

13. Explain in detail about 8259 Programmable Interrupt Controller.

Or

14. Discuss briefly the 8237 DMA Controller with a neat block diagram.

UNIT III

15. Describe the memory mapping and I/O mapping system.

Or

16. Explain in detail the 8279 keyboard and display interfacing.

UNIT IV

17. Discuss the addressing modes of 8086 microprocessor with suitable example.

Or

18. How will you process the interrupts in the 8086 microprocessor?

UNIT V

19. Explain about the various special function registers in 8051 microcontroller and mapping of the Special Function Registers.

Or

20. List and explain the four bi-directional ports of 8051 microcontroller.

5634080

B.Tech. DEGREE EXAMINATION, NOVEMBER 2019.

Fourth Semester

Computer Science and Engineering

MICROPROCESSORS AND MICROCONTROLLERS

(2013–14 onwards)

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. What is Program Counter? Write its use.

2. Expand MAR. What is the use of MAR?

3. Comprehend software and hardware interrupts.

4. State the operating modes of 8253 timer.

5. Compare memory mapped I/O and peripheral mapped I/O.

6. List the operation modes of chip 8255.

7. State the functions of bus interface unit in 8086.

8. What are Assembler Directives? Name any two assembler directives.

9. Discuss the use of TMOD register of 8051 Microcontroller.

10. Name the five interrupts in 8051.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, one questions from each Unit.

All questions carry equal marks.

UNIT I

11. Draw the internal architecture of 8085 Microprocessor and explain each block. (11)

Or

12. (a) Explicate the various addressing modes of 8085 microprocessor with relevant examples. (7)

(b) Write an assembly language program to add an array of numbers stored in memory. (4)

UNIT II

13. Sketch the internal architecture of 8259A and explain its different operating modes. (11)

Or

14. Enumerate the design and working of 8237 DMA Controller with neat block diagram. (11)

UNIT III

15. Deliberate the memory interfacing in 8085. (11)

Or

16. Expound the Port functions of 8255 Programmable Peripheral Interface. (11)

UNIT IV

17. Illustrate the pin description of 8086 with neat pin diagram. (11)

Or

18. Summarize the instruction set of 8086 microprocessor. (11)

UNIT V

19. Draw the internal architecture of 8051 microcontroller and explain each block. (11)

Or

20. Elucidate the special function registers of 8051 with suitable example. (11)

5634080

B.Tech. DEGREE EXAMINATION, MAY 2019.

Fourth Semester

Computer Science Engineering

MICROPROCESSORS AND MICROCONTROLLERS

(2013 – 14 onwards Regulations)

Time : Three hours | Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions

All questions carry equal marks.

1. Specify the size of data, address, memory word and memory capacity of 8085 microprocessor.

2. Mention the purpose of SID and SOD lines.

3. Write the control signals for DMA operation.

4. List the four instructions which control the interrupt structure of the 8085 microprocessor.

5. List the modes of Keyboard.

6. Write the use of 8251 chip.

7. Discuss the function of instruction queue in 8086.

8. What is the maximum memory size that can be addressed by 8086?

9. State the function of RS1 and RS0 bits in the flag register of Intel 8051 microcontroller.

10. Mention the size of DPTR and stack pointer in 8051.

PART B — (5 × 11 = 55 marks)

Answer ONE question from each Unit.

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

11. Draw the pin diagram of 8085 and describe its functions.

Or

12. How to access subroutine from the main program procedure? Illustrate.

13. With a neat sketch, explain 8237 DMAC.

Or

...lain the modes of operation in 8259 ...mmable interrupt controller.

15. Write a program to count number of 1's in the contents of D register and store the count in the B register.

Or

16. Bring out the features of 8251.

17. With diagram, describe the architecture of 8086.

Or

18. Give a brief note on memory addressing.

19. Compare Microprocessor and Microcontroller.

Or

20. How to interface a seven segment display using 8051 microcontroller? Illustrate.

13. With a ...

...plain
...ami...

UNIT V

19. Explain the architecture of 8051 micro controller with a neat diagram. (11)

Or

20. Explain about timing and control instruction set in 8051. (11)

5634080

B.Tech. DEGREE EXAMINATION, NOVEMBER 2018.

Fourth Semester

Computer Science and Engineering

MICROPROCESSORS AND MICROCONTROLLERS

(2013 – 14 onwards batches)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. List the addressing modes of 8085 microprocessor.

2. Define stack.

3. Give the various modes and applications of 8253 programmable interval timer,

4. Draw the block diagram of 8259 programmable interrupt controller.

5. List the major functions performed by CRT interface.

6. Expand USART.

7. What is the purpose of the following commands in 8086

   (a) AAD

   (b) RCL?

8. What is assembler directive?

9. How do you select the register bank in 8051 micro controller?

10. What is interrupt?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain briefly about the internal hardware architecture of 8085 microprocessor with a neat diagram. (11)

Or

12. (a) What are the needs for microprocessors? (6)

    (b) Explain the different types of instruction sets. (5)

UNIT II

13. Explain briefly about Interrupt handling process in 8085. (11)

Or

14. What is DMA? Explain the DMA based data transfer using DMA controller. (11)

UNIT III

15. With the neat diagram explain the 8255 programmable peripheral interface and its operating modes. (11)

Or

16. (a) Explain the different types of memory. (6)

    (b) What is serial communication? Explain. (5)

UNIT IV

17. Explain in detail about the pin description and bus cycles in 8086 microprocessor. (11)

Or

18. (a) Explain the different addressing modes in 8086 microprocessor. (6)

    (b) Explain the types of instruction sets in 8086 microprocessor. (5)

5634080

B.Tech. DEGREE EXAMINATION, MAY 2018.

Fourth Semester

Computer Science

MICROPROCESSORS AND MICROCONTROLLERS

(2013-14 onwards)

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL the questions.

1. List the 16 – bit registers of 8085 microprocessor.
2. Mention the purpose of SID and SOD lines.
3. What is meant by cycle stealing?
4. What is INTR?
5. What is a memory mapped I/O?
6. What is a key debounce?
7. Explain the instructions :
   (a) LDS
   (b) PUSHF
   (c) TEST
   (d) CLD.

8. What is the effect of executing the instruction MOV CX, [SOURCE_MEM]

Where SOURCE_MEM equal to 2016 is a memory location offset relative to the current data segment starting at address $1A000_{16}$.

9. Differentiate between microprocessor and microcontroller.

10. Give the format of PSW register of 8051.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. With suitable examples explain 8085 addressing modes in detail.

Or

12. Write an 8085 Assembly language program to perform 32 bit binary addition.

UNIT II

13. Explain the working of DMA controller with a neat diagram.

Or

14. Explain the interfacing of 8253 with the microprocessor.

UNIT III

15. Sketch and explain the interface of 8279 to the microprocessor. Write an assembly program to read the key codes of keys and display "HELLO".

Or

16. Discuss the organization and architecture of 8251 (USART) with a functional block diagram.

UNIT IV

17. Explain different types of registers in 8086 microprocessor architecture.

Or

18. Write an 8086 program to read a character 'a' from the keyboard.

UNIT V

19. Explain the instruction set of 8051.

Or

20. Write a Program to sort numbers in ascending order using 8051.

5634080

B.Tech. DEGREE EXAMINATION, NOVEMBER 2017.

Fourth Semester

Computer Science and Engineering

MICROPROCESSORS AND MICROCONTROLLERS

(2013 – 14 Onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What do you mean general purpose registers?

2. Why we need a stack in microprocessors?

3. What are the different types of interrupts?

4. List the applications of programmable interval timer.

5. Compare and contrast SRAM and DRAM.

6. What is a RS232C interface?

13.

14. E
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7. What is a flag register?

8. What is an assembler directive?

9. Why we need a special function register?

10. Which of the 8051 port need pull-up registers to function as I/O port?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the Intel 8085 Hardware Architecture with the schematic diagram.

Or

12. What is an addressing mode? Explain all the addressing modes.

UNIT II

13. Why we need a DMA Controller? Explain its hardware architecture with a neat diagram.

Or

14. Draw and explain the block diagram of 8254 Programmable Interval Timer. Also explain the various modes of operation.

UNIT III

15. Explain how to interface key switches and LEDs.

Or

16. With neat block diagram explain the 8255 Programmable Peripheral Interface and its operating modes.

UNIT IV

17. (a) How the interrupt vector is handled in 8086? (6

(b) Draw and explain the timing diagram of write cycle in 8086 in minimum mode. (5

Or

18. (a) Draw the internal block diagram of 8086 microprocessor and explain. (6

(b) Explain the bus cycles of 8086 microprocessor. (5

UNIT V

19. Explain the architecture of 8051 microcontroller with neat diagram.

Or

20. Explain the memory structure of an 8051 Microcontroller.

UNIT V

19. Write short note on

    (a) Internal RAM Storage of 8051 Microcontroller. (6)

    (b) Power control in 8051 Microcontroller. (5)

Or

20. With neat diagram explain the architecture of 8051 Microcontroller with a neat diagram.

5634080

B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017.

Fourth Semester

Computer Science and Engineering

MICROPROCESSOR AND MICROCONTROLLER

(2013 – 2014 onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Define the term 'parity flag'.

2. Mention the six signal groups of 8085 microprocessor.

3. Name the flags that would be affected when performing INR instruction in 8085.

4. What is the role of SP/EN signal in 8259 Programmable Interrupt Controller?

5. State the functionality of RTS and CTS pins in 8251.

6. State the use of CS signal pin in 8255 PPI ?

7. Mention the four segment registers of BIU (Bus Interface Unit) in 8086.

8. Given the effective address of the data m is 341 B and the code segment is 123A. What is the physical address of the data m? (20 bit)

9. Name the special function registers available in 8051.

10. Write down a move instruction that uses 48 as an immediate operand and 58 as a direct address.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE question from each Unit.

All questions carry equal marks.

UNIT I

11. With neat diagram explain the architecture of Intel 8085.

Or

12. Write a 8085 assembly language program for the sum of series of natural numbers (8—bit).

UNIT II

13. Explain the architecture of 8259 programmable Interrupt Controller with a neat diagram.

Or

14. Explain the architecture of 8237 DMA controller with a neat diagram.

UNIT III

15. With neat diagram explain 8279 Keyboard/Display Interface architecture and its functionality.

Or

16. With neat diagram explain 8251 USART for serial data communication.

UNIT IV

17. Describe the various addressing modes of Intel 8086 microprocessor.

Or

18. Explain the minimum and maximum mode operations of Intel 8086.

12.

5634080

B.Tech. DEGREE EXAMINATION,
DECEMBER 2016.

Third Semester

Computer Science and Engineering/
Information Technology

MICROPROCESSORS AND MICROCONTROLLERS

Time : Three hours | Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. What are the functions of an accumulator?
2. What are the different addressing modes of 8085?
3. Define T state and in which T cycle the ALE signal is activated.
4. What is vectored and non-vectored interrupts?
5. How the RS232C serial bus is interfaced to 1TL logic device?
6. What is key debouncing?
7. What is the purpose of segment registers in 8086?

8. State the significance of LOCK signal in 8086.

9. Name the five interrupt sources of 8051.

10. How the program memory is organized in 8051 based system?

PART B — (5 × 11 = 55 marks)

Answer ALL the Questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain different types of instruction set in 8085.

Or

12. Describe the functional block diagram of 8085.

UNIT II

13. Explain the internal architecture of 8237 Direct Memory Access Controller.

Or

14. (a) Explain the mode 0 operation of 8255 programmable peripheral interface. (7)

(b) Explain the different modes of operation of a timer. (4)

UNIT III

15. (a) Draw the block diagram of a keyboard display controller and explain it.

(b) Explain in detail about the parallel communication interface.

Or

16. Write the features of 8251. Discuss how 8251 is used for serial data communication. And also give the advantage of using USART chips in microprocessor based system.

UNIT IV

17. Draw and discuss the interrupt structure of 8086.

Or

18. Write an 8086 ALP to sort out any given 10 numbers in ascending and descending order.

UNIT V

19. With a neat sketch of a schematic diagram explain the functions of various signal of 8051.

Or

20. Write a 8051 program to

(a) Generate Fibonacci series. (6)

(b) Find out the smallest number in an array. (5)

5634080

B.Tech. DEGREE EXAMINATION, APRIL/MAY 2016.

Fourth Semester

Computer Science and Engineering

MICROPROCESSORS AND MICROCONTROLLERS

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Give the purpose of IN and OUT instructions.

2. What is a one byte instruction?

3. List the different types of 8085 interrupts.

4. Give the use of DMA controller.

5. What is Memory Mapped IO?

6. Give the purpose of 8251 USART.

7. What is meant by assembler directive?

8. What is logical memory?

9. What are special function registers?

10. What are ports?

PART B — (5 × 11 = 55 marks)

Answer ALL questions.

All questions carry equal marks.

UNIT I

11. Explain any six addressing modes of 8085 with example.

Or

12. Draw the register organization of 8085. Explain the function of each register and the flags in the flag register.

UNIT II

13. Give the functional description of 8259 Programmable Interrupt Controller.

Or

14. Explain the working of 8253 programmable interval Timer.

UNIT III

15. With example explain I/O interfacing in 8279 Keyboard /Display Interface.

Or

16. Explain the concept of serial communication using RS232C Interface.

UNIT IV

17. Explain the different pins and signals of 8086 microprocessor with a neat diagram.

Or

18. Write an assembly language program to sort 10 numbers.

UNIT V

19. Explain the memory structure of 8051 microcontroller.

Or

20. Explain memory and I/O interfacing in 8051.

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int

UNIT V

19. Draw the architecture of 8051 Microcontroller and explain in detail about them. (11)

Or

20. Write an ALP to find out the smallest number in an array. (11)

5634080

B.Tech. DEGREE EXAMINATION, NOVEMBER 2015.

Fourth Semester

Computer Science and Engineering

MICROPROCESSORS AND MICROCONTROLLERS

(2013-2014 onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is the function of ALE Signal?

2. Write the instructions that are related with stack operation.

3. List out the difference between Hardware and Software Interupts.

4. List out the features of 8259.

5. Compare memory mapping and I/O.

6. What do you mean by Keyboard debouncing?

7. List out the features of Min and Max mode Configuration.

8. What are the functions of Bus Interfacing Unit and Execution Unit?

9. Write the difference between Microprocessor and Microcontroller?

10. What do you mean by TCON, TMOD, SCON and SMOD?

PART B — ($5 \times 11 = 55$ marks).

Answer ALL questions, ONE from each unit.

All questions carry equal marks.

UNIT I

11. Draw the architecture of 8085 and explain in detail about them. (11)

Or

12. Draw the timing diagram for Memory Read and Memory Write operations. (11)

UNIT II

13. Explain in detail about the various modes of operation of 8253. (11)

Or

14. Explain in detail about Working of 8237 with neat diagram. (11)

UNIT III

15. What do you mean by Serial Communication? Also Explain in detail about USART? (11)

Or

16. Explain with interfacing diagram the operation of 8255 with Microprocessor. (11)

UNIT IV

17. Explain in detail about Various Addressing Modes in 8086 with an example. (11)

Or

18. Explain in detail about interrupt processing in 8086. (11)

UNIT IV

17. Explain the architectural features of 8086 microprocessor with its block diagram. (11)

Or

18. (a) Explain the addressing modes of 8086 with examples. (6)

    (b) Discuss the instructions related to shift/Rotate of 8086 microprocessor. (5)

UNIT V

19. With neat diagram, describe in detail about the internal architecture of 8051 microcontroller. (11)

Or

20. (a) Draw and explain the SCON special function register. (5)

    (b) Describe the various interrupts and their associate priorities in 8051 microcontroller. (6)

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5634080

B.Tech. DEGREE EXAMINATION, MAY 2015.

Fourth Semester

Computer Science and Engineering

MICROPROCESSORS AND MICROCONTROLLERS

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Define microprocessor.

2. Give the importance of program counter by 8085 microprocessor.

3. Name the six modes of operation of 8253 programmable interval timer.

4. State the use of cascading signals of 8259 programmable interrupt controller.

5. State the function of DSR and DTR pins in 8251.

6. What do you mean by BSR mode in 8255 PPI?

7. Mention the flags of 8086 processor.

8. What are the min and max mode operations of 8086 processor?

9. What is the size of the program and data memory of 8051 microcontroller?

10. How do you select the register bank in 8051 microcontroller?

PART B — (5 × 11 = 55 marks)

Answer ALL the questions, ONE question from each Unit.

All questions carry equal marks.

UNIT I

11. With neat diagram, explain the architecture of Intel 8085 microprocessor. (11)

Or

12. (a) With suitable examples, explain the addressing modes of 8085 microprocessor. (6)

(b) Draw the timing diagram for memory read machine cycle. (5)

UNIT II

13. (a) Draw the block diagram of interrup[t] controller. (5)

(b) Explain the operation of 8237 DMA controller. (6)

Or

14. (a) List out and explain the maskable and non-maskable interrupts available in 808[5] microprocessor. (5)

(b) Explain the control word format and bit definitions of timer 8253. (6)

UNIT III

15. Draw the block diagram of 8255 and write control word formats in I/O and BSR mode. (11)

Or

16. Explain the functioning of scanned keyboard mode with two key lockout and N key roller of 8279. (11)